

SINAV SORULARI GÖZÜMLERİ

$$1) \lim_{x \rightarrow \frac{\pi}{2}} \frac{\sin x - 1}{\cos^2 x} = \frac{0}{0} \text{ bl.}$$

$$= \lim_{x \rightarrow \frac{\pi}{2}} \frac{\sin x - 1}{\cos^2 x} = \lim_{x \rightarrow \frac{\pi}{2}} \frac{(\sin x - 1) \sin^2 x}{\cos^2 x} \cdot \frac{(\sin x + 1)}{(\sin x + 1)}$$

$$= \lim_{x \rightarrow \frac{\pi}{2}} \frac{(\sin^2 x - 1) \sin^2 x}{\cos^2 x (\sin x + 1)} = \lim_{x \rightarrow \frac{\pi}{2}} \frac{-\cancel{\cos^2 x} \cdot \sin^2 x}{\cancel{\cos^2 x} (\sin x + 1)} = -\frac{1}{2} //$$

$$2) \lim_{x \rightarrow \pi} \frac{\sin(2x)}{x^2 - \pi x} = \frac{0}{0} \text{ bl.}$$

$$= \lim_{x \rightarrow \pi} \frac{2 \cdot \sin x \cdot \cos x}{x(x - \pi)}$$

$$x - \pi = t \Rightarrow x = t + \pi$$

$$x \rightarrow \pi \Rightarrow t = 0$$

$$= \lim_{t \rightarrow 0} \frac{2 \cdot \sin(\pi + t) \cdot \cos(\pi + t)}{(\pi + t) \cdot t}$$

$$\sin(\pi + t) = -\sin t$$

$$\cos(\pi + t) = -\cos t$$

$$= \lim_{t \rightarrow 0} \frac{2 \cdot (-\sin t) \cdot (-\cos t)}{t \cdot (\pi + t)} = \lim_{t \rightarrow 0} \underbrace{\frac{\sin t}{t}}_1 \cdot \underbrace{\frac{2 \cos t}{(\pi + t)}}_{\frac{2}{\pi}} = \frac{2}{\pi} //$$

$$3) \lim_{x \rightarrow \infty} \frac{x - \sqrt{1 + x^2}}{x - \sqrt{x}} \left(\frac{\infty}{\infty} \text{ bl.} \right)$$

$$= \lim_{x \rightarrow \infty} \frac{x - \sqrt{x^2 \left(\frac{1}{x^2} + 1 \right)}}{x \left(1 - \frac{1}{\sqrt{x}} \right)} = \lim_{x \rightarrow \infty} \frac{x - |x| \sqrt{\frac{1}{x^2} + 1}}{x \left(1 - \frac{1}{\sqrt{x}} \right)}$$

$$= \lim_{x \rightarrow \infty} \frac{x \left(1 - \sqrt{\frac{1}{x^2} + 1} \right)}{x \left(1 - \frac{1}{\sqrt{x}} \right)} = \frac{1 - 1}{1} = 0 //$$

$$4) \lim_{x \rightarrow -\infty} (2x + \sqrt{4x^2 + 3x}) = (\infty - \infty \text{ bl.})$$

$$= \lim_{x \rightarrow -\infty} \frac{(2x + \sqrt{4x^2 + 3x})(2x - \sqrt{4x^2 + 3x})}{(2x - \sqrt{4x^2 + 3x})}$$

$$= \lim_{x \rightarrow -\infty} \frac{4x^2 - 4x^2 - 3x}{2x - \sqrt{4x^2 + 3x}} = \lim_{x \rightarrow -\infty} \frac{-3x}{2x - \underset{-x}{|x|} \sqrt{4 + \frac{3}{x}}} = \lim_{x \rightarrow -\infty} \frac{-3x}{2x + x \sqrt{4 + \frac{3}{x^2}}}$$

$$= \lim_{x \rightarrow -\infty} \frac{-3x}{x(2 + \underbrace{\sqrt{4 + \frac{3}{x^2}}}_{\rightarrow 0})} = -\frac{3}{4} //$$

$$5) \lim_{x \rightarrow 0} \frac{1 - \cos(\sin x)}{2x^2} \quad \left(\frac{0}{0} \text{ bl.}\right)$$

$$= \lim_{x \rightarrow 0} \frac{(1 - \cos(\sin x)) \cdot (1 + \cos(\sin x))}{2x^2 (1 + \cos(\sin x))}$$

$$= \lim_{x \rightarrow 0} \frac{1 - \cos^2(\sin x)}{2x^2 (1 + \cos(\sin x))} = \lim_{x \rightarrow 0} \frac{\sin^2(\sin x)}{2x^2 (1 + \cos(\sin x))}$$

$$= \lim_{x \rightarrow 0} \underbrace{\frac{\sin^2(\sin x)}{\sin^2 x}}_1 \cdot \underbrace{\frac{\sin^2 x}{x^2}}_1 \cdot \frac{1}{2(1 + \cos(\sin x))} = \frac{1}{4} //$$

$$6) \lim_{x \rightarrow 2} \frac{|x^2 - 4|}{\sqrt[4]{x - 2}}$$

$$\lim_{x \rightarrow 2^+} \frac{|x^2 - 4|}{\sqrt[4]{x - 2}} = \lim_{x \rightarrow 2^+} \frac{(x-2)(x+2)}{(x-2)^{1/4}} = \lim_{x \rightarrow 2^+} (x-2)^{3/4} (x+2) = 0$$

$$\lim_{x \rightarrow 2^-} \frac{|x^2 - 4|}{\sqrt[4]{x - 2}} = \lim_{x \rightarrow 2^-} \frac{-(x-2)(x+2)}{(x-2)^{1/4}} = \lim_{x \rightarrow 2^-} -(x-2)^{3/4} (x+2) = 0$$

$$\lim_{x \rightarrow 2} \frac{|x^2 - 4|}{\sqrt[4]{x - 2}} = 0 //$$

7) $f(x) = \ln(\ln(\ln x)) + \sqrt{9-x^2}$

$\ln(\ln(\ln x))$
 ≥ 0

$\Downarrow$
 $\ln(\ln x) > 0$

$\Downarrow$
 $\ln x > 1$

$\Downarrow$
 $x > e$

$x > 0$

$\ln x > 0$

$x > 1$

$9-x^2 \geq 0$

x	-3	3
$9-x^2$	-	+

$[-3, 3]$

T.K. : $(e, 3]$

8)

$f(x) = \frac{1}{\sqrt{|x|-x}} + \ln\left(\frac{9-x^2}{x^2+x}\right)$

$|x|-x > 0$

$|x| > x$

$x < 0$

$(-\infty, 0)$

$\frac{9-x^2}{x^2+x} > 0$

$\frac{(3-x)(3+x)}{x(x+1)} > 0$

x	-3	-1	0	3
$9-x^2$	-	+	+	-
x	-	-	+	+
$x+1$	-	-	+	+
$\frac{9-x^2}{x(x+1)}$	-	+	-	+

$(-3, -1) \cup (0, 3)$

$(-\infty, 0) \cap (-3, -1) \cup (0, 3)$

T.K. : $(-3, -1)$

//

9) $f(x) = \frac{3|x-2|}{x^2(4-x^2)}$

$x=0$, $x=\pm 2$ süreksizlik noktaları

$x=0$
 $\lim_{x \rightarrow 0} \frac{3|x-2|}{x^2(2-x)(2+x)} = \lim_{x \rightarrow 0} \frac{3(2-\cancel{x})}{x^2(2-\cancel{x})(2+x)} = \infty$ sonsuz (esas) süreksizdir.

$x=2$

$$\left. \begin{aligned} \lim_{x \rightarrow 2^+} \frac{3|x-2|}{x^2(2-x)(2+x)} &= \lim_{x \rightarrow 2^+} \frac{3(\cancel{x}-2)}{x^2(2-\cancel{x})(2+x)} = -\frac{3}{16} \\ \lim_{x \rightarrow 2^-} \frac{3|x-2|}{x^2(2-x)(2+x)} &= \lim_{x \rightarrow 2^-} \frac{3(2-\cancel{x})}{x^2(2-\cancel{x})(2+x)} = \frac{3}{16} \end{aligned} \right\} \neq \text{sıramalı süreksizdir.}$$

$x=-2$

$\lim_{x \rightarrow -2^+} \frac{3|x-2|}{x^2(2-x)(2+x)} = \lim_{x \rightarrow -2} \frac{3(2-\cancel{x})}{x^2(2-\cancel{x})(2+x)} = \infty$ sonsuz (esas) süreksiz

12)

$f(x) = \frac{2|x-1|}{x^2-x^3}$

$x^2(1-x)=0 \Rightarrow x=0$, $x=1$ süreksizlik noktaları

$x=0$

$\lim_{x \rightarrow 0} \frac{2|x-1|}{x^2(1-x)} = \lim_{x \rightarrow 0} \frac{2(1-\cancel{x})}{x^2(1-\cancel{x})} = \infty$ sonsuz (esas) süreksizlik

$x=1$

$$\left. \begin{aligned} \lim_{x \rightarrow 1^+} \frac{2|x-1|}{x^2(1-x)} &= \lim_{x \rightarrow 1^+} \frac{2(\cancel{x}-1)}{x^2(1-\cancel{x})} = -\frac{2}{1} = -2 \\ \lim_{x \rightarrow 1^-} \frac{2|x-1|}{x^2(1-x)} &= \lim_{x \rightarrow 1^-} \frac{2(1-\cancel{x})}{x^2(1-\cancel{x})} = \frac{2}{1} = 2 \end{aligned} \right\} \neq \text{sıramalı süreksizlik}$$

10)

$$f(x) = \begin{cases} \frac{\pi}{2} & , x=0 \\ \sin\left(\frac{x}{3}\right) & , 0 < x < 3 \\ 2^{\frac{1}{x-4}} & , 3 \leq x < 4 \text{ ve } 4 < x \leq 5 \\ \frac{\pi}{2} & , x=4 \end{cases}$$

 $x=0$

$$\left. \begin{array}{l} \lim_{x \rightarrow 0^+} \sin\left(\frac{x}{3}\right) = 0 \\ f(0) = \frac{\pi}{2} \end{array} \right\} \neq x=0 \text{ da kaldırılabilir süreksizdir.}$$

 $x=3$

$$\left. \begin{array}{l} \lim_{x \rightarrow 3^+} 2^{\frac{1}{x-4}} = \frac{1}{2} = f(3) \\ \lim_{x \rightarrow 3^-} \sin\left(\frac{x}{3}\right) = \sin 1 \end{array} \right\} \neq x=3 \text{ de sıçramalı süreksizdir.}$$

 $x=4$

$$\left. \begin{array}{l} \lim_{x \rightarrow 4^+} 2^{\frac{1}{x-4}} = 2^{\infty} = \infty \\ \lim_{x \rightarrow 4^-} 2^{\frac{1}{x-4}} = 2^{-\infty} = 0 \end{array} \right\} x=4 \text{ de sonsuz (esas) süreksizdir.}$$

11)

$$f(x) = \begin{cases} x & , x \leq -2 \\ 1/(x+2) & , -2 < x \leq 1 \\ \frac{\sin(1-\sqrt{x})}{x-1} & , x > 1 \end{cases}$$

i) $\lim_{x \rightarrow -2} f(x)$, $\lim_{x \rightarrow 1} f(x)$

$$\lim_{x \rightarrow -2^-} x = -2$$

$$\lim_{x \rightarrow -2^+} \frac{1}{x+2} = \infty$$

$\left. \begin{array}{l} \lim_{x \rightarrow -2^-} x = -2 \\ \lim_{x \rightarrow -2^+} \frac{1}{x+2} = \infty \end{array} \right\} \lim_{x \rightarrow -2} f(x) \text{ mevcut değildir.}$

$$\lim_{x \rightarrow 1^-} \frac{1}{x+2} = \frac{1}{3}$$

$$\lim_{x \rightarrow 1^+} \frac{\sin(1-\sqrt{x})}{x-1} = \lim_{x \rightarrow 1^+} \frac{\sin(1-\sqrt{x})}{(\sqrt{x}-1)(\sqrt{x}+1)}$$

$$= \lim_{x \rightarrow 1^+} \frac{\sin(1-\sqrt{x})}{\underbrace{(1-\sqrt{x})}_1} \cdot \frac{-1}{(\sqrt{x}+1)} = -\frac{1}{2}$$

$\left. \begin{array}{l} \lim_{x \rightarrow 1^-} \frac{1}{x+2} = \frac{1}{3} \\ \lim_{x \rightarrow 1^+} \frac{\sin(1-\sqrt{x})}{x-1} = -\frac{1}{2} \end{array} \right\} \neq \text{oldugundan} \\ \lim_{x \rightarrow 1} f(x) \text{ mevcut değildir.}$

ii) $\lim_{x \rightarrow -2^+} \frac{1}{x+2} = \infty$ olduğundan $x = -2$ de sonsuz (esas) süreksiz

$$\lim_{x \rightarrow 1^-} \frac{1}{x+2} = \frac{1}{3} = f(1)$$

$$\lim_{x \rightarrow 1^+} \frac{\sin(1-\sqrt{x})}{x-1} = -\frac{1}{2}$$

$\left. \begin{array}{l} \lim_{x \rightarrow 1^-} \frac{1}{x+2} = \frac{1}{3} = f(1) \\ \lim_{x \rightarrow 1^+} \frac{\sin(1-\sqrt{x})}{x-1} = -\frac{1}{2} \end{array} \right\} \neq \text{oldugundan}$

$x = 1$ de $f(x)$ sıçramalı süreksiz

13) $f(x) = \frac{x^2}{1-\cos x}$ $x=0$ da fonksiyon tanımsız.

$$\lim_{x \rightarrow 0} \frac{x^2}{1-\cos x} = \lim_{x \rightarrow 0} \frac{x^2 (1+\cos x)}{1-\cos^2 x} = \lim_{x \rightarrow 0} \frac{x^2}{\underbrace{\sin^2 x}_1} \cdot (1+\cos x) = 2$$

$x=0$ da $f(x)$ kaldırılabilir süreksizliğe sahiptir.

$$F(x) = \begin{cases} \frac{x^2}{1-\cos x} & , x \neq 0 \\ 2 & , x = 0 \end{cases}$$

14) $f(x) = \frac{|x^2-9|}{x^2-4x+3}$

$x^2-4x+3=0 \Rightarrow (x-3)(x-1)=0 \Rightarrow x=1, x=3$ süreksizlik noktaları

$x=1$
 $\lim_{x \rightarrow 1^+} \frac{|x^2-9|}{(x-1)(x-3)} = \lim_{x \rightarrow 1^+} \frac{-(x-3)(x+3)}{(x-1)(x-3)} = -\infty$ sonsuz (esas) süreksizlik

$x=3$
 $\lim_{x \rightarrow 3^+} \frac{|x^2-9|}{(x-1)(x-3)} = \lim_{x \rightarrow 3^+} \frac{(x-3)(x+3)}{(x-1)(x-3)} = \frac{6}{2} = 3$
 $\lim_{x \rightarrow 3^-} \frac{|x^2-9|}{(x-1)(x-3)} = \lim_{x \rightarrow 3^-} \frac{-(x-3)(x+3)}{(x-1)(x-3)} = -3$
 $\left. \begin{array}{l} \lim_{x \rightarrow 3^+} = 3 \\ \lim_{x \rightarrow 3^-} = -3 \end{array} \right\} \neq \text{sıçramalı süreksizlik}$

15)

$$f(x) = \begin{cases} x^2 \sin \frac{1}{x^2} & , x < 0 \\ a - \arcsin\left(\frac{x+1}{2}\right) & , 0 \leq x < 1 \\ \frac{a}{b} + \arctan \sqrt{3} x & , x \geq 1 \end{cases}$$

$$\lim_{x \rightarrow 0^-} x^2 \sin \frac{1}{x^2} = 0$$

$$\left. \begin{aligned} \lim_{x \rightarrow 0^+} a - \arcsin\left(\frac{x+1}{2}\right) &= a - \frac{\pi}{6} = f(0) \end{aligned} \right\} \Rightarrow a - \frac{\pi}{6} = 0 \Rightarrow \boxed{a = \frac{\pi}{6}} //$$

$$\lim_{x \rightarrow 1^-} \left(a - \arcsin\left(\frac{x+1}{2}\right) \right) = a - \frac{\pi}{2} = \frac{\pi}{6} - \frac{\pi}{2} = -\frac{\pi}{3}$$

$$\lim_{x \rightarrow 1^+} \left(\frac{a}{b} + \arctan \sqrt{3} x \right) = \frac{a}{b} + \frac{\pi}{3} = f(1)$$

$$\left. \begin{aligned} \frac{\pi/6}{b} + \frac{\pi}{3} &= -\frac{\pi}{3} \\ \boxed{b = -\frac{1}{4}} & // \end{aligned} \right\}$$

16) Ara Değer Teoremi: Eğer f $[a, b]$ aralığında sürekli bir fonksiyon ve y_0 da $f(a)$ ve $f(b)$ arasında herhangi bir değer ise bu durumda $y_0 = f(c)$ olacak şekilde (a, b) aralığında bazı c 'ler vardır.

$f(x) = x - \cos x$ olsun. f fonksiyonu $[0, \frac{\pi}{2}]$ aralığında sürekli, $f(0) = -1$ ve $f(\frac{\pi}{2}) = \frac{\pi}{2}$ olduğundan, Ara Değer Teoremine göre,

$$f(0) = -1 < f(c) = 0 < f(\frac{\pi}{2}) = \frac{\pi}{2}$$

$f(c) = 0 = c - \cos c$ olacak şekilde bir $c \in [0, \frac{\pi}{2}]$ vardır.

$$\cos c = c //$$

17)

$$x > 0, f(x) = \frac{1}{\sqrt{x}}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{\frac{1}{\sqrt{x+h}} - \frac{1}{\sqrt{x}}}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\frac{\sqrt{x} - \sqrt{x+h}}{\sqrt{x}\sqrt{x+h}}}{h} = \lim_{h \rightarrow 0} \frac{\sqrt{x} - \sqrt{x+h}}{h \cdot \sqrt{x}\sqrt{x+h}} \cdot \frac{(\sqrt{x} + \sqrt{x+h})}{(\sqrt{x} + \sqrt{x+h})}$$

$$= \lim_{h \rightarrow 0} \frac{x - (x+h)}{h \sqrt{x} \cdot \sqrt{x+h} (\sqrt{x} + \sqrt{x+h})} = \lim_{h \rightarrow 0} \frac{-h}{h \sqrt{x} \sqrt{x+h} (\sqrt{x} + \sqrt{x+h})} \cdot \frac{1}{(\sqrt{x} + \sqrt{x+h})}$$

$$= -\frac{1}{2x\sqrt{x}} = -\frac{1}{2\sqrt{x^3}} //$$

18)

$$f(x) = \begin{cases} \sqrt[3]{x} (1 - \cos x) & , x > 0 \\ \sin x & , x < 0 \end{cases}$$

$$\left. \begin{array}{l} \lim_{x \rightarrow 0^+} \sqrt[3]{x} (1 - \cos x) = 0 = f(0) \\ \lim_{x \rightarrow 0^-} \sin x = 0 \end{array} \right\} \Rightarrow f(x), x=0 \text{ da s\u00fcreklidir.}$$

$$f'_+(0) = \lim_{h \rightarrow 0^+} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0^+} \frac{\sqrt[3]{h} (1 - \cos h)}{h} = 0$$

$$f'_-(0) = \lim_{h \rightarrow 0^-} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0^-} \frac{\sin h}{h} = 1$$

$f'_+(0) \neq f'_-(0)$ olduğundan $f(x), x=0$ da türevlenemezdir.

19) $f(x) = \begin{cases} x^2 + x & , x < 0 \\ x + 1 & , x \geq 0 \end{cases}$

$x=0$ da sürekli mi?

$$\lim_{x \rightarrow 0^+} x + 1 = 1 = f(0)$$

$$\lim_{x \rightarrow 0^-} x^2 + x = 0$$

$\neq$ olduğundan $f(x)$, $x=0$ da sürekli değildir.

Dolayısıyla $x=0$ da türevi mevcut değildir.

20) $f(x) = \begin{cases} \tan(\sin x) & , x \leq 0 \\ \frac{1}{x} \sin x^2 & , x > 0 \end{cases}$

$$\lim_{x \rightarrow 0^+} \frac{1}{x} \sin x^2 = \lim_{x \rightarrow 0^+} \underbrace{\frac{\sin x^2}{x^2}}_1 \cdot x = 0$$

$$\lim_{x \rightarrow 0^-} \tan(\sin x) = 0 = f(0)$$

$\lim_{x \rightarrow 0} f(x) = f(0)$ olduğundan $x=0$ da sürekli dir.

$$f'_+(0) = \lim_{h \rightarrow 0^+} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0^+} \frac{\frac{1}{h} \sin h^2}{h} = \lim_{h \rightarrow 0^+} \frac{\sin h^2}{h^2} = 1$$

$$f'_-(0) = \lim_{h \rightarrow 0^-} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0^-} \frac{\tan(\sin h)}{h} = \lim_{h \rightarrow 0^-} \underbrace{\frac{\tan(\sin h)}{\sin h}}_1 \cdot \underbrace{\frac{\sin h}{h}}_1 = 1$$

$f'_+(0) = f'_-(0) = 1$ olduğundan $f(x)$, $x=0$ de türevlenebilir dir.
 $f'(0) = 1$ //

21)

$$f(x) = \begin{cases} x+1, & x < 0 \\ \cos^2 x, & x \geq 0 \end{cases}$$

$$\left. \begin{array}{l} \lim_{x \rightarrow 0^+} \cos^2 x = 1 = f(0) \\ \lim_{x \rightarrow 0^-} x+1 = 1 \end{array} \right\} = \text{oldugundan } x=0 \text{ da s\u00fcreklidir.}$$

$$f'_+(0) = \lim_{h \rightarrow 0^+} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0^+} \frac{\cos^2 h - 1}{h} = \lim_{h \rightarrow 0^+} \frac{-\sin^2 h \cdot h}{\frac{h^2}{1}} = 0$$

$$f'_-(0) = \lim_{h \rightarrow 0^-} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0^-} \frac{h+1-1}{h} = 1$$

$f'_+(0) \neq f'_-(0)$ oldugundan $x=0$ da t\u00fcrevi mevcut de\u011ildir.

22)

$$g(0) = 8, \quad f(x) = x \cdot g(x)$$

$$f'(0) = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{h g(h) - 0 \cdot g(0)}{h}$$

$$f'(0) = \lim_{h \rightarrow 0} \frac{h g(h)}{h} = g(0) = 8 //$$

23) $0 < x < 1$, $f(x) = \arcsin x - \arccos \sqrt{1-x^2}$

$$f'(x) = \frac{1}{\sqrt{1-x^2}} - \frac{\frac{2x}{2\sqrt{1-x^2}}}{\sqrt{1-(1-x^2)}} = \frac{1}{\sqrt{1-x^2}} - \frac{x}{x\sqrt{1-x^2}} = \frac{1}{\sqrt{1-x^2}} - \frac{1}{\sqrt{1-x^2}} = 0$$

$f'(x) = 0 \Rightarrow f(x) = C$ şeklinde sabit bir fonksiyondur.

24) $f(g(x)) = x$ $f'(x) = 1 + (f(x))^2$

$$(f(g(x)))' = 1$$

$$g'(x) \cdot f'(g(x)) = 1$$

$$g'(x) = \frac{1}{f'(g(x))} = \frac{1}{1 + (f(g(x)))^2} = \frac{1}{1+x^2} //$$

25)

$$x \geq 0, f(x) = \frac{\pi}{4} + \arctan \sqrt{e^{2x} - 1} = \arccos e^{-x}$$

$$f'(x) = \frac{\frac{2e^{2x}}{2\sqrt{e^{2x}-1}}}{1+e^{2x}} - \frac{e^{-x}}{\sqrt{1-e^{-2x}}} = \frac{1}{\sqrt{e^{2x}-1}} - \frac{e^{-x}}{e^{x/2}\sqrt{e^{2x}-1}} = 0$$

$f'(x) = 0 \Rightarrow f(x)$ sabit bir funksiya.

$$x=0 \Rightarrow f(x) = \frac{\pi}{4} //$$

26)

$$f\left(\frac{\pi}{2}\right) = 6, f'\left(\frac{\pi}{2}\right) = 3, g(x) = [f(x)]^{\sin x}$$

$$g(x) = [f(x)]^{\sin x}$$

$$g\left(\frac{\pi}{2}\right) = f\left(\frac{\pi}{2}\right)^{\sin \frac{\pi}{2}} = 6$$

$$\ln g(x) = \sin x \cdot \ln(f(x))$$

$$\frac{g'(x)}{g(x)} = \cos x \cdot \ln(f(x)) + \sin x \cdot \frac{f'(x)}{f(x)}$$

$$\frac{g'\left(\frac{\pi}{2}\right)}{g\left(\frac{\pi}{2}\right)} = \underbrace{\cos \frac{\pi}{2}}_0 \cdot \ln f\left(\frac{\pi}{2}\right) + \underbrace{\sin \frac{\pi}{2}}_1 \cdot \frac{f'\left(\frac{\pi}{2}\right)}{f\left(\frac{\pi}{2}\right)}$$

$$g'\left(\frac{\pi}{2}\right) = 6 \cdot \frac{3}{6} \Rightarrow g'\left(\frac{\pi}{2}\right) = 3 //$$

$$27) \quad x^2 y^2 + \tan(x+y) - 1 = 0, \quad P\left(\frac{\pi}{4}, 0\right)$$

$$\underbrace{2xy^2}_0 + \underbrace{2x^2 y y'}_0 + (1+y')(1+\tan^2(x+y)) = 0$$

$$(1+y'_{lp}) \frac{(1+\tan^2 \frac{\pi}{4})}{2} = 0$$

$$1+y'_{lp} = 0 \Rightarrow \underline{y'_{lp} = -1}$$

$$\text{T.D.D.} : y - y_0 = f'(x_0)(x - x_0)$$

$$y - 0 = -1(x - \frac{\pi}{4})$$

$$y = -x + \frac{\pi}{4} //$$

$$28) \quad y \sin\left(\frac{1}{y}\right) + x \cos\left(\frac{1}{y}\right) = -2x, \quad P\left(0, \frac{1}{\pi}\right)$$

$$\underbrace{y' \sin\left(\frac{1}{y}\right)}_0 - y' \cdot \frac{1}{y} \cos\left(\frac{1}{y}\right) + \cos\left(\frac{1}{y}\right) + x \frac{y'}{y^2} \sin\left(\frac{1}{y}\right) + 2 = 0$$

$$y'_{lp} \pi - 1 + 2 = 0$$

$$y'_{lp} \pi = -1 \Rightarrow \underline{y'_{lp} = -\frac{1}{\pi}}$$

$$\text{T.D.D.} : y - \frac{1}{\pi} = -\frac{1}{\pi}(x - 0)$$

$$y = -\frac{1}{\pi}x + \frac{1}{\pi} //$$

$$29) \quad 2^{xy} + \ln\left[e + \arcsin \frac{y}{x}\right] = 1 + x, \quad P(1, 0)$$

$$\underbrace{(y + xy')}_{\frac{y'x-y}{x^2}} \cdot 2^{xy} \cdot \ln 2 + \frac{\frac{\frac{y'x-y}{x^2}}{\sqrt{1-\frac{y^2}{x^2}}}}{e + \arcsin \frac{y}{x}} = 1$$

$$y'_{lp} \cdot \ln 2 + \frac{y'_{lp}}{e} = 1$$

$$y'_{lp} \left(\ln 2 + \frac{1}{e}\right) = 1$$

$$y'_{lp} = \frac{e}{e \ln 2 + 1}$$

$$\text{T.D.D.} : y - 0 = \frac{e}{e \ln 2 + 1} (x - 1)$$

$$y = \frac{e}{e \ln 2 + 1} x - \frac{e}{e \ln 2 + 1} //$$

$$30) \cos(x-y) = x e^x, \quad P(0, \frac{\pi}{2})$$

$$\cos(x-y) - x e^x = 0$$

$$-(1-y') \cdot \sin(x-y) - e^x - \frac{x e^x}{0} = 0$$

$$(1-y'_{|P}) \cdot \sin \frac{\pi}{2} - 1 = 0$$

$$1-y'_{|P} = 1$$

$$\underline{y'_{|P} = 0}$$

$$\text{T.D.D: } y - \frac{\pi}{2} = 0 \quad (x=0)$$

$$\underline{y = \frac{\pi}{2}} //$$

$$31) \sin(xy) = 1 - x^2 - y^2 + x^2 y^3$$

$$\sin 0 = 1 - x^2 - 0$$

$$1 - x^2 = 0 \Rightarrow \underline{x = \pm 1}$$

$$P_1(1, 0), \quad P_2(-1, 0)$$

$$(y + x y') \cos(xy) = -2x - \frac{2y y'}{0} + \frac{2x y^3}{0} + \frac{3x^2 y^2 y'}{0}$$

$$\underline{P_1(1, 0)}$$

$$m_1 = y'_{|P_1} = -2$$

$$\underline{P_2(-1, 0)}$$

$$m_2 = y'_{|P_2} = -2$$

$m_1 = m_2$ old. dan tegetler birbirine paraleldir.

$$32) y - 2 \cos(\pi y - x) = 2x + 3, \quad P(0, 1)$$

$$y' + 2(\pi y' - 1) \sin(\pi y - x) = 2$$

$$y'_{|P} + 2(\pi y'_{|P} - 1) \sin(\pi - 0) = 2$$

$$\underline{y'_{|P} = 2}$$

$$\text{T.D.D: } y - 1 = 2(x - 0)$$

$$\underline{y = 2x + 1} //$$

$$33) \quad g(1) = h'(1) = g'(1) = h(1) = 2, \quad f(x) = [g(x^2)]^{h(x)}$$

$$f(x) = [g(x^2)]^{h(x)}, \quad f(1) = (g(1))^{h(1)} = 2^2 = 4$$

$$\ln f(x) = h(x) \cdot \ln g(x^2)$$

$$\frac{f'(x)}{f(x)} = h'(x) \cdot \ln g(x^2) + h(x) \cdot \frac{2x g'(x^2)}{g(x^2)}$$

$$\frac{f'(1)}{f(1)} = h'(1) \cdot \ln g(1) + h(1) \cdot \frac{2 g'(1)}{g(1)}$$

$$f'(1) = 4 \cdot \left(2 \cdot \ln 2 + 2 \cdot \frac{2 \cdot 2}{2} \right)$$

$$f'(1) = 8 \ln 2 + 16 //$$

$$35) \quad f(x) = \frac{1+x}{\sqrt{1+x^2}}, \quad f: (-\infty, 1] \rightarrow \mathbb{R}$$

$$f'(x) = \frac{\sqrt{1+x^2} - (1+x) \cdot \frac{1}{2} \cdot 2x (1+x^2)^{-1/2}}{1+x^2}$$

$$f'(x) = \frac{1+x^2 - x \cdot (1+x)}{(1+x^2)^{3/2}} = \frac{1+x^2 - x - x^2}{(1+x^2)^{3/2}} = \frac{1-x}{(1+x^2)^{3/2}} > 0$$

$f'(x) > 0 \Rightarrow f$, 1-1 dir. Dolayısıyla tersi mevcuttur.

$$(f^{-1})'(0) = \frac{1}{f'(f^{-1}(0))}$$

$$\frac{1+x}{\sqrt{1+x^2}} = 0 \Rightarrow x = -1 \quad f'(-1) = \frac{2}{2^{3/2}} = \frac{1}{\sqrt{2}}$$

$$(f^{-1})'(0) = \frac{1}{f'(-1)} = \frac{1}{\frac{1}{\sqrt{2}}} = \sqrt{2} //$$

36) $f(x) = e^{\arctan x}$

$f'(x) = \frac{1}{1+x^2} e^{\arctan x} > 0$ $e^{\arctan x} = e^{\pi/3}$
 $\arctan x = \frac{\pi}{3} \Rightarrow x = \sqrt{3}$

$f, 1-1$ dir ve tersi mevcuttur.

$$(f^{-1})'(e^{\pi/3}) = \frac{1}{f'(f^{-1}(e^{\pi/3}))} = \frac{1}{f'(\sqrt{3})}$$

$$= \frac{1}{1+3} e^{\arctan \sqrt{3}} = \frac{1}{4} e^{\pi/3} //$$

37) $f(x) = 1+x + \ln(1+x^2)$

$f'(x) = 1 + \frac{2x}{1+x^2} = \frac{1+x^2+2x}{1+x^2} = \frac{(1+x)^2}{1+x^2} > 0 \Rightarrow 1-1 \Rightarrow$ tersi var.

$(f^{-1})'(1) = \frac{1}{f'(f^{-1}(1))}$

$$\begin{aligned} 1+x + \ln(1+x^2) &= 1 \\ x + \ln(1+x^2) &= 0 \\ \Rightarrow x &= 0 \\ f'(0) &= \frac{1}{1} = 1 \end{aligned}$$

$$(f^{-1})'(1) = \frac{1}{f'(0)} = \frac{1}{1} = 1 //$$

38) $y = f(x), x=1, 2x+y-1=0$

$2x+y-1=0 \Rightarrow y = -2x+1 \Rightarrow m_N = -2$
 $m_N \cdot m_T = -1 \Rightarrow m_T = \frac{1}{2} = f'(1)$

$x=1 \Rightarrow 2 \cdot 1 + y - 1 = 0$
 $y = -1$

$f(1) = -1$

$$(f^{-1})'(-1) = \frac{1}{f'(1)} = \frac{1}{\frac{1}{2}} = 2 //$$

$$39) f(x) = 2 + \arctan x + e^{2x}$$

$$f'(x) = \frac{1}{1+x^2} + 2e^{2x} > 0 \Rightarrow 1-1 \Rightarrow \text{tersi memcut}$$

$$2 + \arctan x + e^{2x} = 3 \Rightarrow \underline{x=0}, \quad f'(0) = 1+2=3$$

$$(f^{-1})'(3) = \frac{1}{f'(f^{-1}(3))} = \frac{1}{f'(0)} = \frac{1}{3} //$$

$$40) \quad g(t) = t^3 + 7t + 21, \quad g(-2) = -1$$

$$g'(t) = 3t^2 + 7, \quad g^{-1}(-1) = -2$$

$$g^{-1}(t) \approx L(t) = g^{-1}(-1) + (g^{-1})'(-1)(t+1)$$

$$(g^{-1})'(-1) = \frac{1}{g'(g^{-1}(-1))} = \frac{1}{g'(-2)} = \frac{1}{12+7} = \frac{1}{19}$$

$$g^{-1}(t) \approx L(t) = -2 + \frac{1}{19}(t+1) //$$

$$34) \quad f(x) = (\cos x^4)^{\arctan x^2}, \quad f'(0)$$

$$\ln f(x) = \arctan x^2 \ln(\cos x^4), \quad f(0) = 1$$

$$\frac{f'(x)}{f(x)} = \frac{2x}{1+x^4} \ln(\cos x^4) + \arctan x^2 \cdot \frac{-4x^3 \sin x^4}{\cos x^4}$$

$$\frac{f'(0)}{f(0)} = 0 \Rightarrow f'(0) = 0 //$$

$$43) (1,001)^5 - 3(1,001)^{7/3} + 2$$

$$f(x) = x^5 - 3x^{7/3} + 2, \quad a=1,$$

$$f'(x) = 5x^4 - 7x^{4/3}, \quad f(1) = 0, \quad f'(1) = -2$$

$$dy \approx \Delta y \Rightarrow f'(a) dx \approx f(a + \Delta x) - f(a)$$

$$dx = \Delta x = 1,001 - 1 = 0,001$$

$$f'(1) \cdot (0,001) \approx f(1,001) - f(1)$$

$$f(1,001) \approx -0,002 //$$

$$42) g(1) = g'(1) = 4 \quad f(x) = \frac{g(x^2)}{1+x^2}, \quad f(1,25)$$

$$a=1, \quad f(1) = \frac{g(1)}{2} = 2, \quad f'(x) = \frac{2x g'(x^2)(1+x^2) - 2x g(x^2)}{(1+x^2)^2}$$

$$f'(1) = \frac{4g'(1) - 2g(1)}{4}$$

$$f'(1) = \frac{16 - 8}{4} = 2$$

$$f(1,25) \approx L(1,25) = f(1) + f'(1)(1,25 - 1)$$

$$= 2 + 2(0,25) = 2,5$$

$$f(1,25) \approx 2,5 //$$

$$41) f(x) = (x^3 + x - 1)^7 + \arctan(x^4 - 1), \quad x=1, \quad f(1,02)$$

$$f'(x) = 7(x^3 + x - 1)^6(3x^2 + 1) + \frac{4x^3}{1+(x^4-1)^2}$$

$$f(1) = 1, \quad f'(1) = 28 + 4 = 32$$

$$f(1,02) \approx L(1,02) = f(1) + f'(1)(1,02 - 1)$$

$$= 1 + 32 \cdot 0,02 = 1,64$$

$$f(1,02) \approx 1,64 //$$

46) $\sqrt[3]{28}$

$$f(x) = \sqrt[3]{x}, \quad a = 27, \quad f(x) \approx L(x) = f(a) + f'(a)(x-a)$$

$$f'(x) = \frac{1}{3} x^{-2/3} \quad f(27) = 3, \quad f'(27) = \frac{1}{27}$$

$$\begin{aligned} \sqrt[3]{28} = f(28) &\approx L(28) = f(27) + f'(27)(28-27) \\ &= 3 + \frac{1}{27} = 3,037 \end{aligned}$$

$$\sqrt[3]{28} \approx 3,037 //$$

45) $g(2) = -4, \quad g'(x) = \sqrt{x^2+5}, \quad g(2.05)$

$$g(x) \approx L(x) = g(a) + g'(a)(x-a) \quad a = 2, \quad g'(2) = \sqrt{9} = 3$$

$$\begin{aligned} g(2.05) &\approx L(2.05) = g(2) + g'(2)(2.05-2) \\ &= -4 + 3(0,05) \\ &= -4 + 0,15 = -3,85 \end{aligned}$$

$$g(2.05) \approx -3,85 //$$

44) $\sqrt[3]{(63)^2}$

$$f(x) = \sqrt[3]{x^2}, \quad a = 64$$

$$f'(x) = \frac{2}{3} x^{-1/3} \quad f(64) = 16, \quad f'(64) = \frac{1}{6}$$

$$\begin{aligned} f(63) &\approx L(63) = f(64) + f'(64)(63-64) \\ &= 16 + \frac{1}{6}(-1) = 16 - 0,16 = 15,84 \end{aligned}$$

$$\sqrt[3]{(63)^2} \approx 15,84 //$$

47)

$$\lim_{x \rightarrow 0^+} x^{\frac{1}{\ln(e^x - 1)}} = 0^\infty \text{ bl.}$$

$$y = x^{\frac{1}{\ln(e^x - 1)}} \Rightarrow \lim_{x \rightarrow 0^+} \ln y = \lim_{x \rightarrow 0^+} \frac{\ln x}{\ln(e^x - 1)} \quad \left(\frac{\infty}{\infty} \text{ bl.} \right)$$

$$\ln y = \frac{1}{\ln(e^x - 1)} \ln x \Rightarrow \lim_{x \rightarrow 0^+} \ln y = \lim_{x \rightarrow 0^+} \frac{\frac{1}{x}}{\frac{e^x}{e^x - 1}} = \lim_{x \rightarrow 0^+} \frac{e^x - 1}{x e^x} = \frac{0}{0} \text{ bl.}$$

$$\stackrel{\text{L'H}}{=} \lim_{x \rightarrow 0^+} \frac{e^x}{e^x + x e^x} = 1$$

$$\lim_{x \rightarrow 0^+} \ln y = \ln \lim_{x \rightarrow 0^+} y = 1$$

$$\Rightarrow \lim_{x \rightarrow 0^+} x^{\frac{1}{\ln(e^x - 1)}} = e^1 = e //$$

48)

$$\lim_{x \rightarrow \infty} [1 + 2^x + 3^x]^{\frac{1}{x}} = \infty^\circ \text{ bl.}$$

$$\lim_{x \rightarrow \infty} [1 + 2^x + 3^x]^{\frac{1}{x}} = \lim_{x \rightarrow \infty} \left[3^x \left(\frac{1}{3^x} + \left(\frac{2}{3} \right)^x + 1 \right) \right]^{\frac{1}{x}}$$

$$= \lim_{x \rightarrow \infty} 3 \cdot \left[\underbrace{\frac{1}{3^x}}_{\rightarrow 0} + \underbrace{\left(\frac{2}{3} \right)^x}_{\rightarrow 0} + 1 \right]^{\frac{1}{x}}$$

$$= 3 \cdot 1 = 3 //$$

49)

$$\lim_{x \rightarrow 4} \left[\frac{4}{\pi} \arctan\left(\frac{x}{4}\right) \right]^{\tan\left(\frac{\pi x}{8}\right)} = 1^\infty \text{ bl.}$$

$$y = \left[\frac{4}{\pi} \arctan\left(\frac{x}{4}\right) \right]^{\tan\left(\frac{\pi x}{8}\right)}$$

$$\ln y = \tan\left(\frac{\pi x}{8}\right) \cdot \ln\left(\frac{4}{\pi} \arctan\frac{x}{4}\right)$$

$$\lim_{x \rightarrow 4} \ln y = \lim_{x \rightarrow 4} \frac{\ln\left(\frac{4}{\pi} \cdot \arctan\frac{x}{4}\right)}{\cot\left(\frac{\pi x}{8}\right)} = \frac{0}{0} \text{ bl.}$$

$$\stackrel{\text{L'H}}{=} \lim_{x \rightarrow 4} \frac{\cancel{4} \cdot \frac{1/4}{1 + \frac{x^2}{16}}}{\frac{4}{\pi} \cdot \arctan\frac{x}{4}} \cdot \frac{1}{-\frac{\pi}{8} \cdot \operatorname{cosec}^2\left(\frac{\pi x}{8}\right)}$$

$\parallel$
 $\sin^2\left(\frac{\pi x}{8}\right)$

$$= \lim_{x \rightarrow 4} \frac{-8 \sin^2\left(\frac{\pi x}{8}\right)}{4\pi\left(1 + \frac{x^2}{16}\right) \cdot \arctan\frac{x}{4}} = \frac{-8}{4\pi \cdot 2 \cdot \frac{\pi}{4}} = -\frac{4}{\pi^2}$$

$$\lim_{x \rightarrow 4} \ln y = \ln \lim_{x \rightarrow 4} y = -\frac{4}{\pi^2}$$

$$\lim_{x \rightarrow 4} y = e^{-\frac{4}{\pi^2}} //$$

50)

$$\lim_{x \rightarrow 0^+} (2 - e^{\sqrt{x}})^{\frac{1}{x}} \quad (1^\infty \text{ bl.})$$

$$y = (2 - e^{\sqrt{x}})^{\frac{1}{x}}$$

$$\ln y = \frac{1}{x} \ln(2 - e^{\sqrt{x}})$$

$$\lim_{x \rightarrow 0^+} \ln y = \lim_{x \rightarrow 0^+} \frac{\ln(2 - e^{\sqrt{x}})}{x} = \left(\frac{0}{0} \text{ bl.}\right)$$

$$\stackrel{\text{L'H}}{=} \lim_{x \rightarrow 0^+} \frac{-\frac{1}{2\sqrt{x}} \cdot e^{\sqrt{x}}}{2 - e^{\sqrt{x}}} = -\infty$$

$$\lim_{x \rightarrow 0^+} \ln y = \ln \left(\lim_{x \rightarrow 0^+} y \right) = -\infty$$

$$\Rightarrow \lim_{x \rightarrow 0^+} (2 - e^{\sqrt{x}})^{\frac{1}{x}} = e^{-\infty} = 0 //$$

51)

$$\lim_{x \rightarrow 0^+} (\cot x)^{1/\ln x} \quad (\infty^\infty \text{ bl.})$$

$$y = (\cot x)^{1/\ln x}$$

$$\ln y = \frac{1}{\ln x} \cdot \ln \cot x$$

$$\lim_{x \rightarrow 0^+} \ln y = \lim_{x \rightarrow 0^+} \frac{\ln \cot x}{\ln x} \stackrel{\infty/\infty}{=} \stackrel{\text{L'H}}{=} \lim_{x \rightarrow 0^+} \frac{\frac{-1/\sin^2 x}{\cot x}}{\frac{1}{x}} = \lim_{x \rightarrow 0^+} \frac{-x \cdot \sin x}{\sin^2 x \cdot \cos x}$$

$$= \lim_{x \rightarrow 0^+} \frac{-x}{\sin x \cdot \cos x} = \lim_{x \rightarrow 0^+} \underbrace{\frac{x}{\sin x}}_1 \cdot \frac{-1}{\cos x} = -1$$

$$\lim_{x \rightarrow 0^+} \ln y = \ln \left(\lim_{x \rightarrow 0^+} y \right) = -1$$

$$\Rightarrow \lim_{x \rightarrow 0^+} (\cot x)^{1/\ln x} = e^{-1} = \frac{1}{e} //$$

52)

$$\lim_{x \rightarrow \infty} \frac{e^{\arctan x} - x}{\ln(1+x^2) + x} \quad \left(\frac{\infty}{\infty} \text{ bl.} \right) \stackrel{\text{L'H}}{=} \lim_{x \rightarrow \infty} \frac{\frac{1}{1+x^2} \cdot e^{\arctan x} - 1}{\frac{2x}{1+x^2} + 1}$$

$$= \frac{0 \cdot e^{\pi/2} - 1}{0 + 1} = -1 //$$

53)

$$\lim_{x \rightarrow 0^+} \left(\frac{a^x + b^x}{2} \right)^{\frac{2}{x}} \quad (1^\infty \text{ bl.})$$

$$y = \left(\frac{a^x + b^x}{2} \right)^{\frac{2}{x}}$$

$$\ln y = \frac{2}{x} \ln \left(\frac{a^x + b^x}{2} \right)$$

$$\lim_{x \rightarrow 0^+} \ln y = \lim_{x \rightarrow 0^+} \frac{\ln \left(\frac{a^x + b^x}{2} \right)}{\frac{x}{2}} \quad \left(\frac{0}{0} \text{ bl.} \right) \stackrel{\text{L'H}}{=} \lim_{x \rightarrow 0^+} \frac{2 \cdot \frac{a^x \ln a + b^x \ln b}{2(a^x + b^x)}}{\frac{1}{2}}$$

$$= \lim_{x \rightarrow 0^+} \frac{2(a^x \ln a + b^x \ln b)}{a^x + b^x} = \ln a + \ln b$$

$$\lim_{x \rightarrow 0^+} \ln y = \ln \left(\lim_{x \rightarrow 0^+} y \right) = \ln(a \cdot b)$$

$$\Rightarrow \lim_{x \rightarrow 0^+} \left(\frac{a^x + b^x}{2} \right)^{\frac{2}{x}} = a \cdot b //$$

54)

$$\lim_{x \rightarrow 1^+} \left[\frac{4}{\pi} \arctan x \right]^{\frac{3}{x^2+2x-3}} \quad (1^\infty \text{ bl.})$$

$$y = \left[\frac{4}{\pi} \arctan x \right]^{\frac{3}{x^2+2x-3}}$$

$$\ln y = \frac{3}{(x^2+2x-3)} \ln \left(\frac{4}{\pi} \arctan x \right)$$

$$\lim_{x \rightarrow 1^+} \ln y = \lim_{x \rightarrow 1^+} \frac{3 \cdot \ln \left(\frac{4}{\pi} \arctan x \right)}{(x^2+2x-3)} \quad \left(\frac{0}{0} \text{ bl.} \right)$$

$$\stackrel{\text{L'H}}{=} \lim_{x \rightarrow 1^+} \frac{3 \cdot \frac{\frac{4}{\pi} \cdot \frac{1}{1+x^2}}{\frac{4}{\pi} \arctan x}}{2x+2} = \lim_{x \rightarrow 1^+} \frac{3}{(2x+2)(1+x^2) \arctan x}$$

$$\lim_{x \rightarrow 1^+} \ln y = \frac{3}{4 \cdot 2 \cdot \frac{\pi}{4}} = \frac{3}{2\pi}$$

$$\ln \left(\lim_{x \rightarrow 1^+} y \right) = \frac{3}{2\pi} \Rightarrow \lim_{x \rightarrow 1^+} \left[\frac{4}{\pi} \arctan x \right]^{\frac{3}{x^2+2x-3}} = e^{\frac{3}{2\pi}} //$$

55)

$$\lim_{x \rightarrow 0^+} (1 + \sin 2x)^{\frac{1}{x}} \quad (1^\infty \text{ bl.})$$

$$y = (1 + \sin 2x)^{\frac{1}{x}}$$

$$\ln y = \frac{1}{x} \ln(1 + 2x)$$

$$\lim_{x \rightarrow 0^+} \ln y = \lim_{x \rightarrow 0^+} \frac{\ln(1+2x)}{x} \quad \left(\frac{0}{0} \text{ bl.} \right)$$

$$\stackrel{\text{L'H}}{=} \lim_{x \rightarrow 0^+} \frac{\frac{2}{1+2x}}{1} = 2$$

$$\lim_{x \rightarrow 0^+} \ln y = 2$$

$$\ln \left(\lim_{x \rightarrow 0^+} y \right) = 2 \Rightarrow \lim_{x \rightarrow 0^+} (1 + \sin 2x)^{\frac{1}{x}} = e^2 //$$